LESSON PLAN

AUXILIARY EQUIPMENT (18)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	06
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Concept of reduction gear
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Concept of reduction gear
5. Specific Objectives	18.1.2.1 Trainee will be able to explain functioning of gear, their classification

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of gear diagrams from IB; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a gear train?
(b) How does gear ratio affect output speed?
(c) Why are reduction gears used in aircraft engines?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of reduction gear • Working principle of gear reduction • Input and output shaft relationship • Torque and speed transformation	L & Ds	PPT, IB	15 min
(b)	• Classification of reduction gears • Simple vs compound gear trains • Epicyclic gear systems • Helical and spur gear types	L & Ds	PPT, PDF	20 min
(c)	• Efficiency considerations in gears • Heat generation and lubrication • Common failure modes • Maintenance implications	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how a reduction gear increases torque.
(b) Differentiate between simple and compound gear trains.
(c) List two common failure modes in reduction gears.

12. Home Assignment and Follow Up	Prepare a brief note on the application of reduction gears in the next lesson topic (LP 07 - Operation & functions of bearing) focusing on how bearing loads relate to gear output torque.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Operation & functions of bearing.	Refer to the master syllabus reference list	LP 07

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min